

PAPER_246: MUGE Dual-Mode Oscillatory Gravity — Standing Wave and Hubble-Normalised Traveling Wave

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Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — MUGEDualModeOscillatoryGravityCalculator (Session 62, grokshare8d951e12.txt 4th-pass) **Date:** March 2026 **Series:** Phase 2 Session 62 — §3.x Universal MUGE Sub-Term Integration

$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \text{Bigl}(1 - [SSq] \cdot U_{bi}, \wedge, F_U(r,t) \text{Bigr}, \quad [SSq] = 0.57$$

Abstract

Gravity in the MUGE framework is not a static field — it supports oscillatory modes that arise from the interference of inward- and outward-propagating gravitational perturbations. This paper establishes the **dual-mode oscillatory gravity sub-term** (g_{osc}), which is the superposition of two distinct wave modes: a standing wave and a Hubble-normalised traveling wave.

Mode 1 (standing wave): $g_{\text{osc1}} = 2A \cdot \cos(kx) \cdot \cos(\omega t)$ — the classic interference pattern of two equal-amplitude counter-propagating waves. Mode 2 (traveling wave): $g_{\text{osc2}} = (2p/T_{\text{H_gyr}}) \cdot A \cdot \cos(kx - \omega t)$ — a unidirectional propagating disturbance whose amplitude is suppressed by the inverse Hubble time in gigayears, $(2p/T_{\text{H_gyr}})$, connecting gravitational oscillations to the cosmological expansion rate.

The key resonance condition — $\omega_{\text{local}} = 2p/t_{\text{Hubble}}$ — places the system at the threshold where Mode 2 dominates the superposition because $(2p/T_{\text{H_gyr}}) \approx 1$ for $T_{\text{H_gyr}} = 2p$. Away from resonance, Mode 1 dominates for Hubble times much larger than $2p$ Gyr. The time-averaged result $\langle g_{\text{osc}} \rangle = 0$ ensures no net secular drift — oscillatory gravity is a zero-mean perturbation to the static MUGE field.

UQFF Discovery: Novel application of UQFF calibration constants ($\omega = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. System Parameters and Equation Overview

Parameter	Symbol	Default Value	Units	Meaning
Oscillation amplitude	A_{osc}	1×10^{-10}	m/s^2	Gravitational wave amplitude

Parameter	Symbol	Default Value	Units	Meaning
Wavenumber	k	1/r	1/m	Spatial frequency at system scale r
Angular frequency	?	2pc/r	rad/s	Relativistic frequency at scale r
Hubble time	tH _{gyr}	13.8	Gyr	Current epoch value
Position	x	variable	m	Spatial coordinate
Time	t	variable	s	Note: in context, passed as epoch

Primary equations:

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Mode 1 (standing wave):
g_osc1 = 2 · A · cos(k·x) · cos(·t)
Mode 2 (Hubble-normalised traveling wave):
g_osc2 = (2p / T_H_gyr) · A · cos(k·x - ·t)
Total:
g_osc = g_osc1 + g_osc2
Time average:
?g_osc? = 0 (both modes are zero-mean over integer wave periods)

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Mode 2 amplitude factor at TH_{gyr} = 13.8:

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(2p / 13.8) ~ 0.455

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2. Core Physics Derivation

2.1 Standing Wave — Counter-Propagating Superposition

A standing gravitational wave arises when two plane waves with equal amplitude A, wavenumber k, and frequency ? travel in opposite directions along x:

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g? = A · cos(kx - ?t) [forward]
g? = A · cos(kx + ?t) [backward]
g_standing = g? + g? = 2A · cos(kx) · cos(?t) [trig identity]

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This mode is spatially modulated by cos(kx) — nodes at $kx = (n+\frac{1}{2})p$, antinodes at $kx = np$. At nodes, gravity is unaffected by the standing wave; at antinodes, the oscillation reaches full amplitude 2A.

2.2 Traveling Wave — Hubble-Time Amplitude Suppression

Mode 2 is a single traveling wave whose amplitude is modulated by a cosmological suppression factor:

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g_osc2 = (2p / T_H_gyr) · A · cos(k·x - ·t)

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The factor $(2p / T_H_gyr)$ has units of 1/Gyr when TH_{gyr} is in gigayears, but since A is already in m/s², the result is dimensionally consistent only if TH_{gyr} is treated as a dimensionless ratio $\tau_H / (1 \text{ Gyr})$. This convention is standard in cosmological normalisation within MUGE.

Physical interpretation: Gravitational disturbances traveling at cosmological speeds are attenuated by the Hubble expansion. The factor $(2p/T_H_gyr)$ is the instantaneous angular expansion rate in units of (Gyr)⁻¹, analogous to the Hubble parameter $H_0 = 1/t_{\text{Hubble}}$ but expressed in the natural angular frequency unit.

2.3 Resonance Condition

At resonance, the traveling-wave amplitude equals the standing-wave amplitude:

$$(2p / T_{H_gyr}) = 1 \Rightarrow T_{H_gyr} = 2p \sim 6.28 \text{ Gyr}$$

For the current Universe ($T_{H_gyr} = 13.8$), Mode 2 amplitude factor ~ 0.455 — Mode 2 carries about 45% of Mode 1 amplitude. At early cosmic times ($T_{H_gyr} \approx 2p \sim 6.3 \text{ Gyr}$, i.e., $z \sim 0.5$ in Λ CDM), the two modes were equal in amplitude. At Mode 2 resonance ($T_{H_gyr} = 2p$), the interference pattern is maximally complex.

2.4 Zero Time Average

Both sinusoidal modes average to zero over a complete oscillation period:

$$\begin{aligned} \langle \cos(kx) \cdot \cos(\omega t) \rangle_t &= \cos(kx) \cdot \langle \cos(\omega t) \rangle_t = 0 \\ \langle \cos(kx - \omega t) \rangle_t &= 0 \\ \langle g_{osc} \rangle_t &= 0 \end{aligned}$$

This result ensures that oscillatory gravity is a **perturbative zero-mean correction** to the static MUGE field — it modulates gravity on timescale $2p/\omega = r/c$ (light-crossing time of the system) but produces no secular drift in the total gravitational potential.

2.5 Wavenumber and Frequency at Astrophysical Scales

For a system of radius r , the natural wavenumber and frequency are $k = 1/r$ and $\omega = 2pc/r$. This choice ties the oscillation period to the light-crossing time:

$$T_{osc} = 2p/\omega = r/c$$

For $r = 1 \text{ kpc}$: $T_{osc} \sim 3.3 \text{ kyr}$ — *much shorter than stellar evolution timescales, so the oscillation averages out over physical processes.* For $r = 1 \text{ Mpc}$: $T_{osc} \sim 3.3 \text{ Myr}$ — comparable to galaxy cluster merger timescales.

3. Dual-Mode Zero-Mean Theorem

Theorem (MUGE Oscillatory Zero Mean): The dual-mode oscillatory sub-term $g_{osc} = g_{osc1} + g_{osc2}$ is a zero-mean bounded perturbation to the static MUGE field for all systems with finite r . The maximum instantaneous amplitude is $|g_{osc}|_{max} = A \cdot (2 + 2p/T_{H_gyr})$, reached when both modes constructively interfere at an antinode. No secular modification of total MUGE gravity results from this term in the time-averaged limit.

The **Hubble resonance condition** $T_{H_gyr} = 2p$ is the unique epoch at which Mode 1 and Mode 2 have equal amplitude, producing the most complex gravitational interference pattern observable.

4. Observational Predictions / Validation

Gravitational wave background: The dual-mode structure predicts a specific spatial correlation pattern in the stochastic gravitational wave background — standing-wave nodes should appear as directions of suppressed GW strain in future pulsar timing arrays (IPTA, SKA).

Galaxy cluster mass oscillations: At $r \sim \text{Mpc}$, $T_{osc} \sim 3 \text{ Myr}$ — oscillatory gravity contributes to ICM pressure waves seen in Chandra X-ray maps. The mode-2/mode-1 amplitude ratio (0.455 at $z=0$) is a direct probe of the Hubble constant at the cluster.

Early Universe enhancement: At $z \sim 0.5$ ($TH_{\text{gyr}} \sim 2p$), the standing and traveling waves were equal — enhanced gravitational perturbations at this epoch may leave an imprint in the large-scale galaxy power spectrum at $k \sim 0.1 \text{ h/Mpc}$.

5. References

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